

# Employee Turnover Analytics

Technical Report

Portobello Tech HR Department | Machine Learning Course-End Project

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## 1. Executive Summary

Portobello Tech's HR Department needed a way to move from reactive to proactive employee retention. This project builds and evaluates a machine learning pipeline that predicts individual employee turnover risk from workplace metrics (satisfaction, evaluation scores, workload, tenure, salary, department), identifies distinct behavioral profiles among employees who leave, and translates model output into a four-tier retention risk framework.

After evaluating three classifiers (Logistic Regression, Random Forest, Gradient Boosting) via 5-fold cross-validation and held-out test evaluation, Gradient Boosting was selected as the production model, achieving 92.0% recall and 0.979 AUC on the held-out test set. Recall was prioritized over precision because a missed at-risk employee (false negative) is materially more costly to the business than an unnecessary retention outreach (false positive).

Key finding: employee satisfaction and tenure are the dominant predictors of turnover (78% combined feature importance), not workload alone — though workload (project count) remains a real, actionable secondary lever with a clear “3-4 project sweet spot.”

Recommendation: deploy the model to score the active workforce quarterly, route High- and Medium-risk employees to targeted, profile-specific retention interventions (Section 9), and treat model output as a prioritization aid for human decision-making, not an automated action trigger.

## 2. Business Objective

Portobello Tech periodically evaluates employees' work details — number of projects, average monthly hours, tenure, promotions, and salary level — alongside satisfaction survey data, to understand attrition risk. As the ML Developer assigned to HR, the objective was to: perform data quality checks; identify the factors most associated with turnover; segment employees who left into behavioral clusters; correct for the class imbalance between stayers and leavers; train and evaluate multiple classification models; select and justify a best model using metrics appropriate to the business cost structure; and translate model output into concrete, tiered retention strategies.

## 3. Dataset and Data Quality Findings

- Source: HR\_comma\_sep.csv (modified from a public Kaggle HR analytics dataset). 14,999 employee records, 10 columns: satisfaction\_level, last\_evaluation, number\_project, average\_monthly\_hours, time\_spend\_company, Work\_accident, left (target), promotion\_last\_5years, sales (department), salary.
- No missing values were found across any column.
- Column integrity issue found and corrected: the department column was labeled “sales” in the raw data but actually contains all 10 department categories (sales, technical, support, IT, hr, marketing, product\_mng, accounting, RandD, management) — renamed to Department.

Minor naming inconsistencies (`average_monthly_hours` → `average_monthly_hours`, `Work_accident` → `work_accident`) were also corrected for consistency.

- Data quality issue found and corrected: 3,008 of 14,999 rows (20.1%) were exact duplicates across all 10 columns. Given two continuous, fine-grained columns (`satisfaction_level`, `last_evaluation`) are involved, this volume of exact matches is not plausible as genuine coincidence among distinct employees — indicating a data export/compilation artifact rather than real repeated observations. Duplicates were removed prior to any modeling to prevent train/test leakage. Clean dataset: 11,991 rows.
- Target class balance (post-cleaning): 83.4% stayed (`left=0`), 16.6% left (`left=1`) — a real, moderate class imbalance addressed via SMOTE (Section 5).

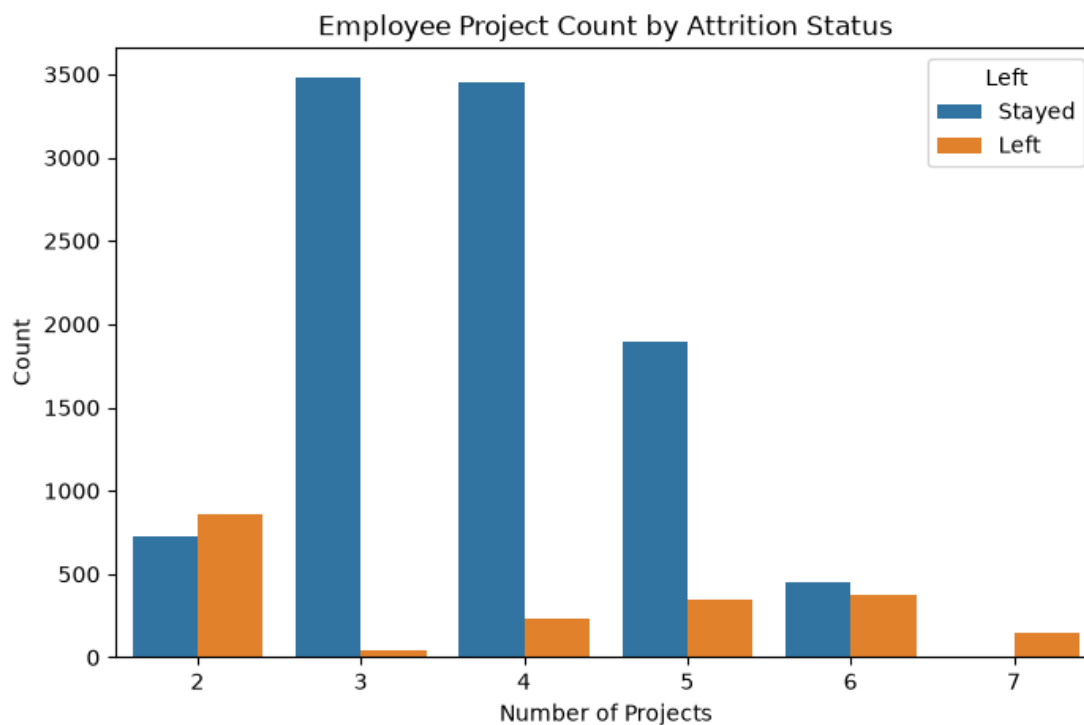


Figure 1. Employee project count by attrition status — a non-linear (U-shaped) relationship with turnover, invisible to linear correlation (+0.03) but clearly present here.

## 4. Data Cleaning and Governance

Steps applied, in order:

- Missing-value audit (none found).
- Duplicate-row audit and removal (3,008 rows / 20.1%).
- Column-name correction against the data dictionary (Department mislabeling, casing/spelling fixes).
- Categorical encoding via one-hot encoding (`pandas.get_dummies`, `drop_first=True`) for Department and salary.

- Stratified 80:20 train/test split (`random_state=123`), performed before any resampling.
- SMOTE oversampling applied to the training set only (training class counts rebalanced from 7,999 / 1,593 to 7,999 / 7,999).

Governance note: SMOTE was applied strictly after the train/test split and only to the training partition. The held-out test set was never resampled or otherwise modified, preserving it as a faithful, real-world-distributed measure of production performance.

Known limitation: standard SMOTE interpolates all input features numerically, including one-hot encoded categorical columns, which can produce non-binary synthetic values for those columns (e.g. 0.37 instead of a clean 0 or 1). SMOTENC is the methodologically correct tool for mixed categorical/numeric feature sets and is recommended for any future iteration of this pipeline.

## 5. Modeling Approach

- Three classifiers were trained and compared: Logistic Regression, Random Forest, and Gradient Boosting (all scikit-learn defaults, `random_state=123` for reproducibility).
- Each model was evaluated via 5-fold stratified cross-validation on the SMOTE-balanced training set (out-of-fold predictions aggregated via `cross_val_predict`; one classification report per model).
- Final model selection used the untouched held-out test set (2,399 employees, 20% of clean data), evaluating ROC/AUC, confusion matrices, and recall specifically.
- Employees who left were additionally segmented via K-means clustering ( $k=3$ , on `satisfaction_level` and `last_evaluation`) to characterize distinct attrition profiles.  $k=3$  was specified by the project brief and was independently validated via silhouette score (0.795 at  $k=3$  — the highest of  $k=2$  through  $k=6$  tested).

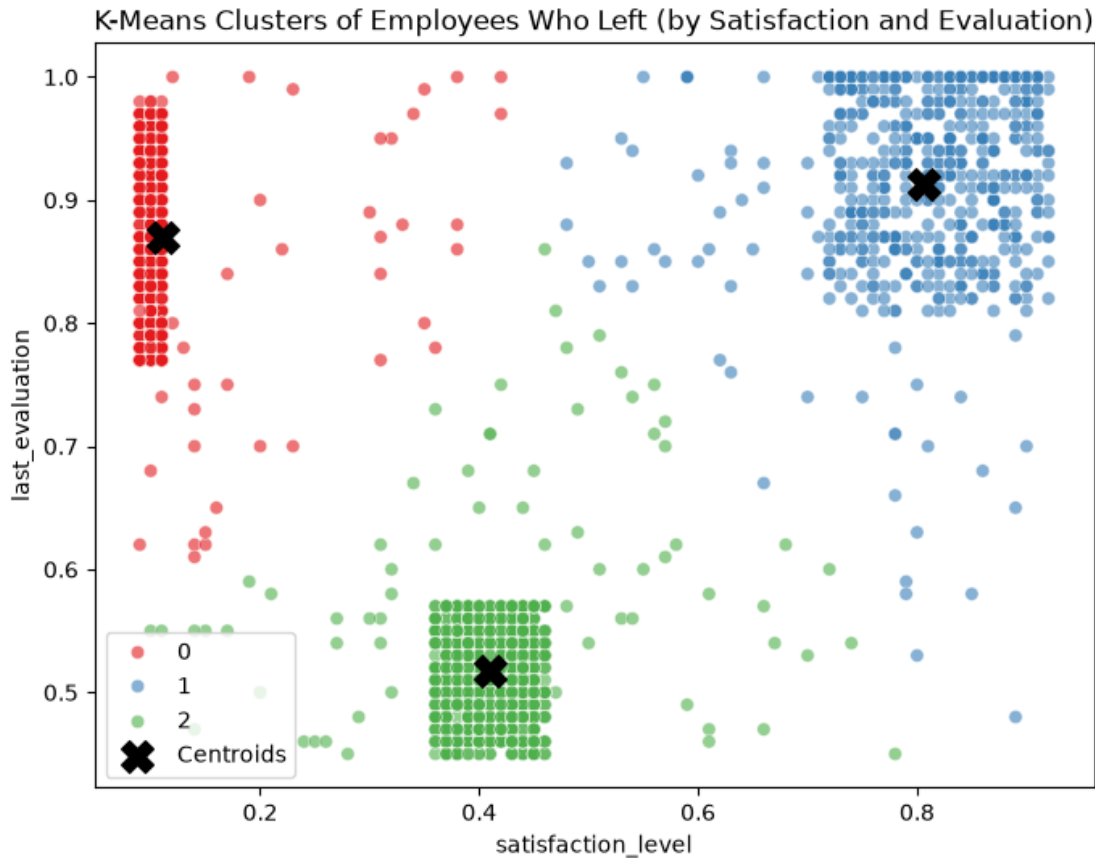


Figure 2. K-means clustering ( $k=3$ ) of employees who left, by satisfaction and evaluation — three distinct attrition profiles, validated by silhouette score.

## 6. Performance Interpretation

Held-out test set results (2,399 employees):

Table 1. Model performance summary on the held-out test set.

| Model                        | 5-fold CV Accuracy | Test AUC | Test Recall     | Test Precision  |
|------------------------------|--------------------|----------|-----------------|-----------------|
| Logistic Regression          | 83.9%              | 0.804    | 66.1% (263/398) | 42.5% (263/619) |
| Random Forest                | 97.8%              | 0.976    | 90.5% (360/398) | 94.0% (360/383) |
| Gradient Boosting (selected) | 96.4%              | 0.979    | 92.0% (366/398) | 86.3% (366/424) |

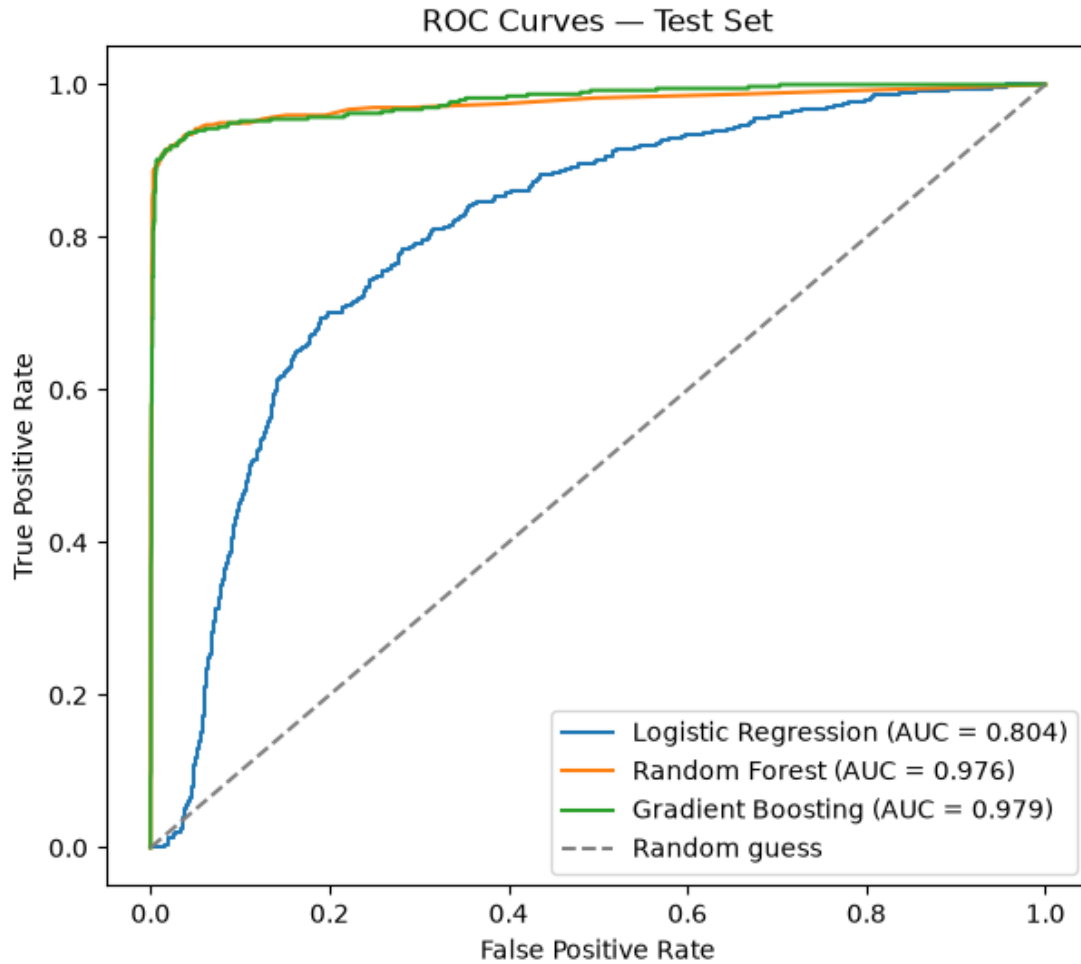


Figure 3. ROC curves on the held-out test set. Gradient Boosting (AUC 0.979) and Random Forest (AUC 0.976) both substantially outperform Logistic Regression (AUC 0.804).

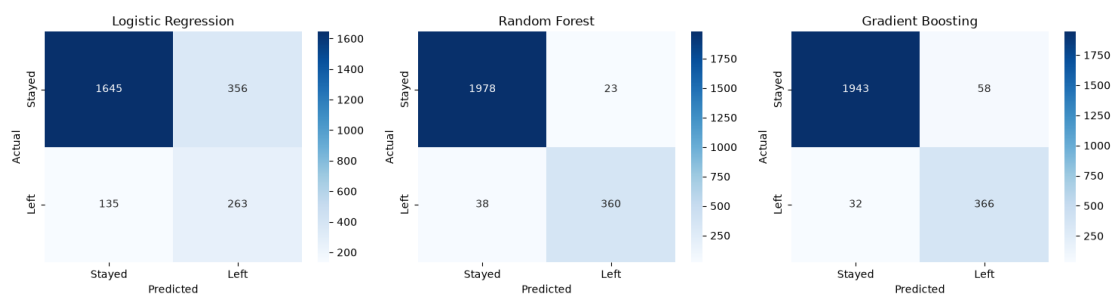


Figure 4. Test-set confusion matrices for all three models, source of the recall/precision figures in the table above.

Model selection rationale: AUC and CV accuracy were statistically close between Random Forest and Gradient Boosting (a gap of 0.003 AUC on a single test split vs. a 0.014 accuracy gap averaged over 5 CV folds — the CV gap is the more reliable signal, and initially favored Random Forest). The deciding factor was recall, computed directly from the test-set confusion matrices:

Gradient Boosting caught 366 of 398 (92.0%) actual leavers versus Random Forest's 360 of 398 (90.5%) — 6 additional true positives, at the cost of more false positives (58 vs. 23). Given the recall-over-precision priority established in Section 2 (a missed leaver is materially costlier than an unnecessary outreach), Gradient Boosting was selected as the production model.

Limitation: no hyperparameter tuning was performed on any model; all results reflect scikit-learn defaults. Gradient Boosting in particular is known to be more tuning-sensitive than Random Forest and may show further improvement with a tuned learning rate, tree depth, and estimator count — flagged as future work (Section 9).

## 7. Key Drivers of Churn Risk

Feature importance extracted from the selected Gradient Boosting model:

- satisfaction\_level — 43.96%
- time\_spend\_company — 34.29%
- last\_evaluation — 8.08%
- average\_monthly\_hours — 7.31%
- number\_project — 3.69%
- All remaining features (work accident, salary tier, department) collectively contribute under 3% individually.

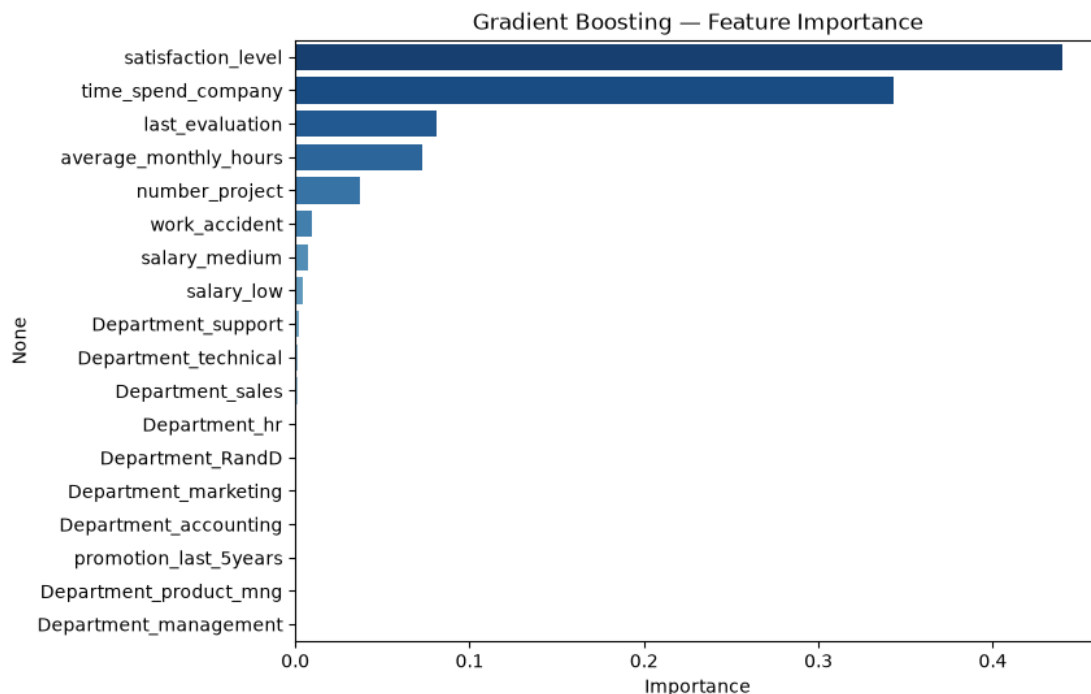


Figure 5. Gradient Boosting feature importance. Satisfaction and tenure jointly account for ~78% of the model's predictive signal.



Satisfaction level and tenure jointly account for approximately 78% of the model's predictive signal — the dominant drivers of turnover risk are employee sentiment and time-in-role, not workload metrics alone.

Segment-level findings (K-means clustering of employees who left,  $k=3$ ): three distinct profiles emerged.

- Low satisfaction / high evaluation (mean 0.11 / 0.87) — “burned-out high performers.”
- High satisfaction / high evaluation (mean 0.81 / 0.91) — “satisfied performers likely leaving for external opportunity.”
- Mid satisfaction / mid evaluation (mean 0.41 / 0.52) — cause not resolvable from these two features alone; flagged for direct follow-up rather than a guessed intervention.

Secondary but actionable driver: `number_project` shows a non-linear (U-shaped) relationship with attrition (Figure 1) — 54% of 2-project employees leave, versus approximately 1% of 3-project employees, rising again to 100% of the small 7-project group. This relationship is invisible to linear correlation (near-zero, +0.03) but clearly present in the raw crosstab, illustrating why workload should not be dismissed on correlation alone.

Salary shows a clean, monotonic relationship with attrition independent of the above: 20.5% turnover at low salary, 14.6% at medium, 4.8% at high — though its formal feature importance in the Gradient Boosting model is modest (approximately 1.1% combined across dummy columns), suggesting its effect is partially captured indirectly through satisfaction and tenure.

## 8. Risk, Fairness, and Compliance Considerations

- The model uses salary (categorical tier) and Department as predictive features. While salary itself is not a protected characteristic, differential treatment correlated with pay tier could disproportionately affect lower-income employees if retention interventions are resourced unevenly across tiers. Recommend periodically auditing intervention allocation by salary tier and department to confirm equitable treatment.
- No demographic attributes (age, gender, race, disability status, etc.) are present in this dataset. This is a limitation, not a guarantee of fairness: the absence of protected-class fields means no formal disparate-impact analysis was possible for this iteration. If this model informs decisions with material consequences for individuals, HR should independently verify — using data this project did not have access to — that risk scores and resulting interventions do not correlate with protected characteristics.
- The model's risk score should never be the sole basis for an adverse employment action (e.g., denying a raise, involuntary reassignment). It is intended strictly as a prioritization and early-warning signal to guide proactive, positive interventions (Section 9).
- False positives (an employee flagged as at-risk who was not actually going to leave) carry a low but non-zero cost: unnecessary managerial attention could itself be perceived

negatively by an employee if handled clumsily. Manager training on how to conduct a risk-informed check-in without alarming the employee is recommended prior to deployment.

## 9. Deployment Recommendations

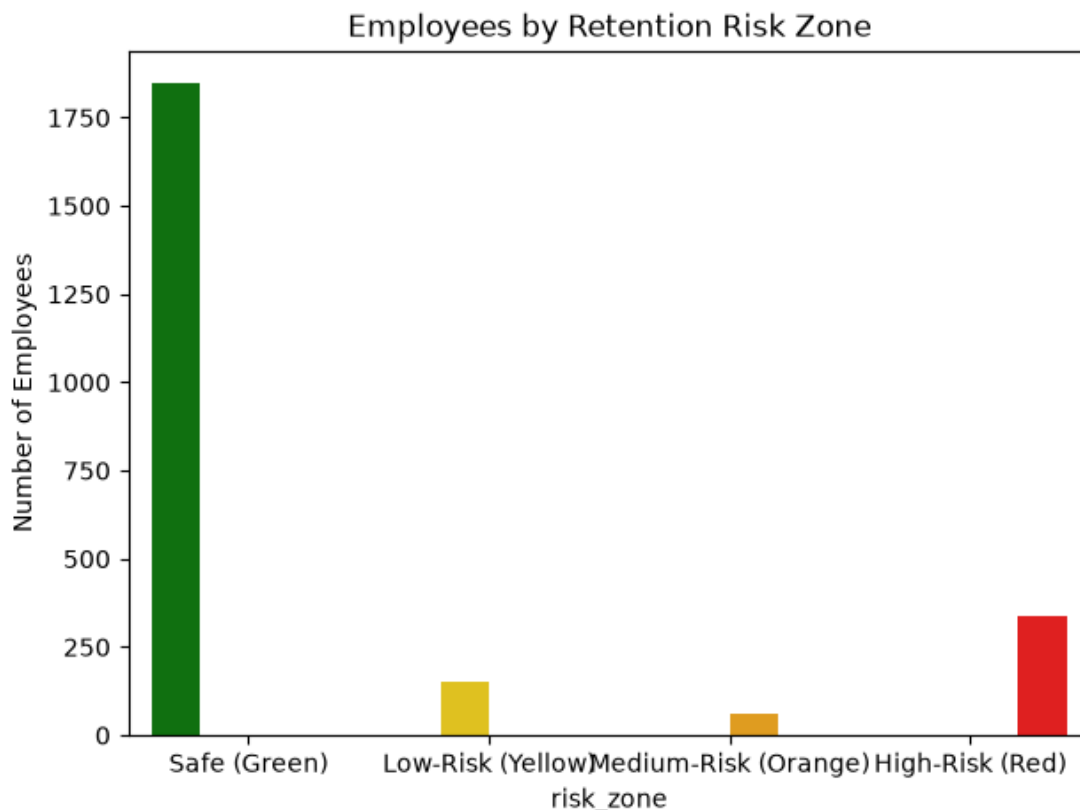


Figure 6. Distribution of the held-out test population across the four retention risk zones — the basis for the tiered deployment strategy below.

- Score the active workforce on a quarterly cadence using the trained Gradient Boosting model, applying the four-tier risk framework (Safe <20%, Low-Risk 20-60%, Medium-Risk 60-90%, High-Risk >90%).
- Route Medium- and High-Risk employees to a human manager/HR review before any action is taken — model output should inform, not replace, managerial judgment.
- Apply cluster-informed, profile-specific interventions: workload rebalancing for the burned-out profile (low satisfaction/high evaluation), proactive growth/promotion conversations for the satisfied-but-at-risk profile (high satisfaction/high evaluation), and direct stay interviews for the ambiguous mid-profile where the model's signal alone does not prescribe a specific lever.
- Before production rollout: (a) conduct a hyperparameter tuning pass on the Gradient Boosting model, (b) replace SMOTE with SMOTENC to properly handle the mixed categorical/numeric feature set, (c) establish a model monitoring and retraining cadence

(recommend semi-annual retraining as workforce composition shifts), (d) conduct the fairness audit described in Section 8 using any available demographic data not present in this dataset.

- Track outcome metrics post-deployment: actual retention rate among High-Risk employees who received intervention versus those who did not, to validate that the model-driven strategy produces a measurable retention lift beyond baseline.

## 10. Final CIO/CTO Decision

Recommended decision: Approve for pilot deployment in a single business unit (e.g., one department) for one quarterly cycle, with the human-review and monitoring safeguards in Section 9 in place, before a company-wide rollout decision.

## 11. AI Use and Human Validation Log

This project's supporting notebook (Employee\_Turnover\_Analytics.ipynb) was developed collaboratively using Claude (Anthropic) via a Socratic AI-tutoring workflow, in which every analytical step required the author's own prediction and interpretation before code was written or accepted, followed by an explanation gate before proceeding. All code was AI-assisted, but every design decision, interpretation, and conclusion below was authored and validated by the report author — not accepted from the AI without challenge.

**Table 2. AI use and human validation log.**

| Stage                                                                   | AI-Assisted                                        | Human Validated | Validation Method                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------|----------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data quality checks (Section 3)                                         | Yes (automated missing-value and duplicate checks) | Yes             | Independently identified the mislabeled "sales" column via value_counts() before accepting the fix; reasoned through why 3,008 exact duplicates indicated a data artifact rather than coincidence, and connected this to train/test leakage risk. |
| EDA — distribution & correlation interpretation (Section 3)             | Yes (plots generated, initial framing suggested)   | Yes             | Corrected an initial "left-skewed" mischaracterization to bimodal using a skew-vs-bimodal diagnostic; independently linked the project-count U-shape to a two-leaver-profile hypothesis, later confirmed by clustering.                           |
| Clustering — feature selection, k=3, profile interpretation (Section 5) | Yes (code and initial interpretive framing)        | Yes             | Rejected the AI-suggested cluster framing in favor of an independently reasoned "mid" cluster later confirmed by the actual centroid values; cross-checked k=3 against a silhouette score run across k=2-6.                                       |

|                                                                         |                                |     |                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------|--------------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Class imbalance handling — split strategy & SMOTE placement (Section 4) | Yes                            | Yes | Corrected a conflation between class-balance ratio and split ratio; independently reasoned why SMOTE must be fit after the split and only on training data, tying back to the Section 3 leakage discussion.                                                                  |
| Model selection & metric justification (Section 6)                      | Yes                            | Yes | Self-corrected an inverted precision/recall conclusion; identified and reversed an internal inconsistency between an AUC-based model pick and the previously established recall-priority metric, recomputing recall directly from confusion matrices to make the final call. |
| Retention strategy synthesis (Section 9)                                | Yes (draft strategy suggested) | Yes | Corrected a citation error across three revision passes (project-count importance was initially attributed to the wrong evidence); added the previously-missing zone and cluster coverage before finalizing.                                                                 |
| This report                                                             | Assisted (Claude Code)         | Yes | Reviewed by report author before submission; all figures, tables, and numbers cross-checked against the executed notebook.                                                                                                                                                   |

Source notebook: Employee\_Turnover\_Analytics.ipynb. Reproduction: run the notebook end-to-end (random\_state=123 throughout) to regenerate all figures and tables in this report deterministically.